BOILER PROCESS MONITORING & CONTROL SY

Using Siemens TIA Portal V17 and Ladder Logic

Engineering Project Dossier & Educational Reference Model

PROJECT METADATA

Platform: Siemens TIA Portal V17 | SIMATIC S7-1200 | S7-PLCSIM V17

Language: Ladder Diagram [LAD / KOP] - IEC 61131-3

Target Controller: CPU 1214C DC/DC/DC - 6ES7 214-1AG40-0XB0 - FW V4.5

Scope: Deterministic Boiler Drum Level Monitoring & Alarm Annunciation

Status: Complete Engineering Specification & Simulation Model

EXECUTIVE ABSTRACT

This project establishes an educational, simulated PLC-based boiler drum water-level monitoring and alarm system engineered in Siemens TIA Portal V17.

A simulated drum water level variable [Water_Level, REAL, 0.0% to 100.0%] is cyclically compared against a configurable low-level trip threshold [Low_Level_Threshold, REAL, 30.0%] using a Siemens CMP < comparator instruction in Main [OB1].

When the water level drops strictly below the threshold, a non-latching Boolean alarm [Low_Level_Alarm, BOOL] activates. The alarm automatically clears upon level recovery.

A supervisory diagnostic tier flags out-of-range sensor and setpoint values. The full design is verified through a 10-point test matrix in S7-PLCSIM with Watch Tables.

1. PLC TAG ARCHITECTURE & MEMORY MAPPING

The controller utilizes symbolic addressing mapped to deterministic S7-1200 memory offsets:

Tag Name	Address	Type	Initial Value	Functional Purpose
Water_Level	%MD10	Real	60.0	Simulated drum water level [0.0% - 100.0%]
Low_Level_Threshold	%MD14	Real	30.0	Configurable low water alarm trip setpoint
Low_Level_Alarm	%M0.0	Bool	FALSE	Discrete low water alarm output indicator
Water_Level_Invalid	%M0.1	Bool	FALSE	[Extended] Diagnostic flag for level out-of-bounds
Threshold_Invalid	%M0.2	Bool	FALSE	[Extended] Diagnostic flag for threshold out-of-bounds

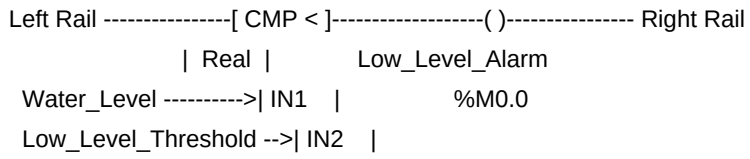
Critical Memory Rule: 4-Byte Allocation for REAL Variables

A REAL data type occupies 32 bits [4 contiguous bytes] conforming to IEEE 754.
%MD10 spans bytes %MB10, %MB11, %MB12, %MB13. The next available address is %MD14.
Allocating %MD12 would overlap bytes 12 and 13, causing severe data corruption.

2. LADDER DIAGRAM [LAD] NETWORK SPECIFICATION

Network 1: Low Water-Level Detection & Alarm Assignment

Ladder Rung Representation:



Evaluation Mechanics:

- When Water_Level < Low_Level_Threshold evaluates to TRUE, power flow passes to coil.
- The coil -( )- writes 1 [TRUE] to %M0.0.
- When Water_Level >= Low_Level_Threshold evaluates to FALSE, power flow is interrupted.
- The standard coil -( )- writes 0 [FALSE] to %M0.0, automatically clearing the alarm.

3. VERIFICATION TEST MATRIX & RESULTS

The logic is verified across 10 deterministic test cases in Siemens S7-PLCSIM:

Test ID	Water_Level	Threshold	Logic Expression	Expected Alarm	Behavior Category
TC-01	80.0%	30.0%	80.0 < 30.0 -> FALSE	FALSE	High Normal Operating Level
TC-02	60.0%	30.0%	60.0 < 30.0 -> FALSE	FALSE	Mid-Range Operating Level
TC-03	30.0%	30.0%	30.0 < 30.0 -> FALSE	FALSE	Boundary: Level == Threshold
TC-04	29.9%	30.0%	29.9 < 30.0 -> TRUE	TRUE	Boundary: 0.1% Sub-threshold
TC-05	20.0%	30.0%	20.0 < 30.0 -> TRUE	TRUE	Standard Low-Level Trip
TC-06	0.0%	30.0%	0.0 < 30.0 -> TRUE	TRUE	Dry Drum Condition
TC-07	45.0%	30.0%	45.0 < 30.0 -> FALSE	FALSE	Recovery Test [Auto-clears]
TC-08	25.0%	40.0%	25.0 < 40.0 -> TRUE	TRUE	Dynamic Setpoint Adjustment
TC-09	50.0%	40.0%	50.0 < 40.0 -> FALSE	FALSE	Dynamic Setpoint Safe Level
TC-10	20.0%	20.0%	20.0 < 20.0 -> FALSE	FALSE	Boundary: Setpoint == Level

4. INTERVIEW HIGHLIGHTS & RESUME INTEGRITY

Key Interview Defense Points:

1. Why REAL instead of INT? Analog process values are continuous decimals. INT would truncate 29.9% to 29 or 30, eliminating sub-percentage precision near critical limits.
2. Equal-to-threshold behavior: CMP < strictly evaluates inequality. 30.0 < 30.0 is FALSE.
3. Non-latching vs Latching: Advisory alarms automatically clear when condition clears. Emergency shutdown trips [which cut fuel] must latch until manual operator reset.
4. Scope & Honesty: This project is an educational PLC simulation. It does not claim physical sensor wiring or certified safety-instrumented system [SIS] deployment.